

Model Evaluation Report — Personalized Math Tutor

(3-Way Comparison)

2026-07-20 | 20 held-out problems | Greedy decoding (temperature=0.2, do_sample=False) | SymPy-verified grading

Training Telemetry (Loss & Meta)

Model	Eval Loss	Epochs	Mean Token Acc %
base	N/A (Base)	N/A	N/A
v1	0.6289	3.0	84.1%
v2	1.201	2.0	73.36%
v3	0.6445	15.0	84.08%
v4	0.5279	4.7272727272727275	86.17%

Overall Summary

Model	Overall Accuracy (%)	Unparseable Count	Avg Response Length
base	55.0%	0	553.2 chars
v1	40.0%	0	673.1 chars
v2	50.0%	0	429.8 chars
v3	55.0%	0	712.8 chars
v4	60.0%	0	653.5 chars

Epoch vs Accuracy Correlation

Model	Epochs Trained	Math Accuracy (%)	Unparseable Count
base	N/A (Base)	55.0%	0
v1	3.0	40.0%	0
v2	2.0	50.0%	0
v3	15.0	55.0%	0
v4	4.73	60.0%	0

Per-Topic Accuracy Breakdown (%)

Model	Linear Equations	Quadratic Factoring	Ratio	Systems	Probability	Statistics	Exponential
base	66.7%	33.3%	33.3%	0.0%	100.0%	100.0%	50.0%
v1	66.7%	66.7%	0.0%	0.0%	66.7%	33.3%	50.0%
v2	100.0%	66.7%	0.0%	66.7%	66.7%	0.0%	50.0%
v3	66.7%	100.0%	0.0%	0.0%	33.3%	100.0%	100.0%
v4	33.3%	66.7%	0.0%	33.3%	100.0%	100.0%	100.0%

Comparative Analysis Summary

- **Highest overall accuracy:** v4 at 60.0%.
- **Lowest overall accuracy:** v1 at 40.0%.
- **v2 vs v1 accuracy delta:** +10.0% (v2 improved over v1).
- **v1 vs base:** -15.0% (drop over the base model).
- **v2 vs base:** -5.0% (drop over the base model).

Per-Topic Best / Worst Model

Topic	Best Model	Best Acc (%)	Worst Model	Worst Acc (%)
Linear Equations	v2	100.0%	v4	33.3%
Quadratic Factoring	v3	100.0%	base	33.3%
Ratio	base	33.3%	v4	0.0%
Systems	v2	66.7%	v3	0.0%
Probability	base	100.0%	v3	33.3%
Statistics	base	100.0%	v2	0.0%
Exponential	v3	100.0%	v2	50.0%